Practical 08 & 09

Practical - 08

To configure wireless network and Implementation of IoT using Cisco Packet tracer

Contd…

Aim:

A. To configure a wireless network using Cisco Packet Tracer.

B. To implement an IoT setup in Cisco Packet Tracer.

Objective:

A. To configure a network using a wireless router and a few end devices.

B. Configuring IoT network with smart devices.

Simulation Tool: Cisco Packet Tracer

Procedure: a. Wireless Networks

- Start up Cisco Packet Tracer, and drag a wireless router into the workspace.
- Select a few end devices and drag them onto the workspace as well.
- Double click on the router, and click on the GUI tab to begin configuration.
- Set a name for your network, and select WEP type security. Set a password.
- Double click on each of the devices, and in their physical view, turn off the power, before replacing any one of the modules with the WPC300N module.

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- Next, click on PC Wireless in the Desktop tab, then go to the ‘Connect’ tab.
- The wireless network of your chosen name should soon appear in the list. Select it, enter the password you set earlier, then click on Connect.
- After you have done this for all end devices, select any one, go to Command Prompt, and attempt to ping any other device (the IP address will automatically be set thanks to DHCP). If the ping succeeds, the network has been set up.

Procedure: b. IoT

- Start up Cisco Packet Tracer, and select a few Home devices.
- Select a Home Gateway (Wireless Network Device) and drag it onto the workspace.
- Go to 'Wireless' in its Settings after double clicking it, and copy its SSID.
- Double click each device, go to 'Config' and select Home Gateway as the IoT server.
- Finally, select a Smartphone, double click, go to 'Config' and paste the SSID.
- You will now see all devices connected to the Gateway. Go to IoT Monitor in Desktop if you'd like to manipulate your chosen end devices through your Smartphone.

Practical - 09

Imagine two processes communicate across a network. One process running in your local system is web browser and the other process running in the remote system is the web server. Once a connection is established between the web browser and web server, the server's current date and time has to be displayed in web browser. Write a suitable program for this scenario (Python).

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Aim:

To develop a TCP client-server program in which a client (web browser) establishes a connection with a remote server and displays the server's current date and time.

Objective:

To implement TCP-based client-server communication for retrieving and displaying the server's current date and time at the client side.